Presentations with Beamer

LaTeX: A Professional Document Preparation System

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Agenda

1. The `beamer` Package
2. Adding Effects
3. Formatting Columns
4. Highlighting Stuffs
5. Using Themes and colorthemes

Introduction

Here is a working example of a beamer presentation.

```
\documentclass{beamer}

\title{The Beamer Package}
\author{John Doe}
\institute{A Famous University}
\date{25 December 2023}

\begin{document}

\begin{frame}
\maketitle
\end{frame}

\begin{frame}
\frametitle{Introduction}
This is a slide. You can put some contents here.
For example, \[ x^2 + g(y) = \int f(x) \mathrm{d}x + y^3. \]
\end{frame}

\end{document}
```

The title page

```
\title{The Beamer Package}
\subtitle{Creating good looking presentations in LaTeX}

\author{John Doe\inst{1} and Sam Willy\inst{2}}

\institute{
\inst{1}%
Faculty of Mathematics\\
A Famous University
\and
\inst{2}%
Faculty of Computer Science\\
Another Famous University
}

\date{25 December 2023}

\logo{\includegraphics[height=2cm]{ctan_lion.png}}
```

Example

The Beamer Package

Creating good looking presentations in LaTeX

John Doe[1] and Sam Willy[2]

[1]Faculty of Mathematics
A Famous University

[2]Faculty of Computer Science
Another Famous University

25 December 2023



...e contents here. For example,

$$\int f(x)\mathrm{d}x + y^3.$$



Creating a table of contents

```
\begin{frame}
\frametitle{Table of Contents}
\tableofcontents
\end{frame}

\section{Introduction}
\begin{frame}
\frametitle{Introduction}
This is a slide. You can put some contents
 ↪  here.
For example, \[ x^2 + g(y) = \int f(x)
 ↪  \mathrm{d}x + y^3. \]
\end{frame}
```

Example

Table of Contents

Introduction

Recent Works

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Analysis & Discussions

Conclusion



Adding effects to items

```
\begin{frame}
\frametitle{Sample frame title}
This is a text in second frame.
For the sake of showing an example.

\begin{itemize}
 \item<1- > Text visible from slide 1.
 \item<2-3> Text visible from slide 2 to slide 3.
 \item<3> Text visible only on slide 3.
 \item<4- > Text visible from slide 4.
\end{itemize}
\end{frame}
```

Example

This is a text in second frame. For the sake of showing an example.

- Text visible from slide 1.

Example

This is a text in second frame. For the sake of showing an example.

- Text visible from slide 1.
- Text visible from slide 2 to slide 3.

Example

This is a text in second frame. For the sake of showing an example.

- Text visible from slide 1.
- Text visible from slide 2 to slide 3.
- Text visible only on slide 3.

Example

This is a text in second frame. For the sake of showing an example.

- Text visible from slide 1.

- Text visible from slide 4.

Adding effects to a paragraph

```
\begin{frame}
\onslide<1- >{%
Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Curabitur non pellentesque leo. }

\onslide<2- >{%
\[ x^2 + y^2 = z^2 \] }

\onslide<3>{%
Phasellus porttitor malesuada augue,
sit amet congue neque pharetra ac.
Fusce maximus orci non viverra ultricies.
Morbi nec condimentum turpis.
Nullam eleifend varius mauris eget fringilla.}

\onslide<4>{%
Integer finibus suscipit est et molestie.}
\end{frame}
```

Example

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur non pellentesque leo.

Example

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur non pellentesque leo.

$$x^2 + y^2 = z^2$$

Example

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur non pellentesque leo.

$$x^2 + y^2 = z^2$$

Phasellus porttitor malesuada augue, sit amet congue neque pharetra ac. Fusce maximus orci non viverra ultricies. Morbi nec condimentum turpis. Nullam eleifend varius mauris eget fringilla.

Example

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur non pellentesque leo.

$$x^2 + y^2 = z^2$$

Integer finibus suscipit est et molestie.

Adding effects with pauses

```
\begin{frame}
Lorem ipsum dolor sit amet,
consec adipisc elit. Curabit pellentes leo. \pause

\[ x^2 + y^2 = z^2 \] \pause

Phasellus porttitor malesuada augue,
sit amet congue neque pharetra ac.
Fusce maximus orci non viverra ultricies. \pause
Morbi nec condimentum turpis.
Nullam eleifend varius mauris fringilla. \pause

Integer finibus suscipit est et molestie.
\end{frame}
```

Example

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur non pellentesque leo.

Example

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur non pellentesque leo.

$$x^2 + y^2 = z^2$$

Example

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur non pellentesque leo.

$$x^2 + y^2 = z^2$$

Phasellus porttitor malesuada augue, sit amet congue neque pharetra ac. Fusce maximus orci non viverra ultricies.

Example

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur non pellentesque leo.

$$x^2 + y^2 = z^2$$

Phasellus porttitor malesuada augue, sit amet congue neque pharetra ac. Fusce maximus orci non viverra ultricies. Morbi nec condimentum turpis. Nullam eleifend varius mauris eget fringilla.

Example

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur non pellentesque leo.

$$x^2 + y^2 = z^2$$

Phasellus porttitor malesuada augue, sit amet congue neque pharetra ac. Fusce maximus orci non viverra ultricies. Morbi nec condimentum turpis. Nullam eleifend varius mauris eget fringilla.
Integer finibus suscipit est et molestie.

Columns

Sometimes the information in a presentation looks better in a two-column format:

```
\begin{frame}
\frametitle{Two-column slide}
\begin{columns}
\column{0.5\textwidth}
This is a text in first column.
$$E=mc^2$$
\begin{itemize}
\item First item
\item Second item
\end{itemize}

\column{0.5\textwidth}
This text will be in the second column
and on a second thoughts, this is a nice looking
layout in some cases.
\end{columns}
\end{frame}
```

Example

This is a text in first column.

$$E = mc^2$$

- First item
- Second item

This text will be in the second column and on a second thoughts, this is a nice looking layout in some cases.

Highlighting important stuffs

A good practice is to highlight the important points to make it easier for audiences to identify the main topic.

```
\begin{frame}
\frametitle{Sample frame title}

In this slide, some important text will be
\alert{highlighted} because it's important.
Please, don't abuse it.

\begin{block}{Remark}
Sample text
\end{block}

\begin{alertblock}{Important box}
Sample text in red box
\end{alertblock}

\begin{examples}
Sample text in green box. The title of the block is
↪  ``Examples".
\end{examples}
\end{frame}
```

Example

In this slide, some important text will be highlighted because it's important. Please, don't abuse it.

Remark

Sample text

Important box

Sample text in red box

Examples

Sample text in green box. The title of the block is "Examples".

Using Themes

The Madrid theme is set by adding the following command:

```
\documentclass{beamer}
\usetheme{Madrid}

\title{The Beamer Package}
\subtitle{Creating good looking presentations in LaTeX}
\author{John Doe\inst{1} and Sam Willy\inst{2}}
\institute{
\inst{1}%
Faculty of Mathematics\\A Famous University
\and
\inst{2}%
Faculty of Computer Science\\Another Famous University
}
\date{25 December 2023}
\logo{\includegraphics[height=2cm]{ctan_lion.png}}
\begin{document}
\begin{frame}
\maketitle
\end{frame}
```

Beamer Theme: Madrid

The Beamer Package

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Beamer Theme: Madrid

Table of Contents

1. Introduction
2. Recent Works
3. Main Results
4. Analysis & Discussions
5. Conclusion



Beamer Theme: Madrid

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1 Intro
2 Rece
3 Main
4 Anal
5 Conc

John Doe and Sa

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John Doe and Sa

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Remark

Sample text

Important box

Sample text in red box

Examples

Sample text in green box. The title of the block is "Examples".



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Beamer Theme: Berkeley



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Beamer Theme: Berkeley



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1 Introduction

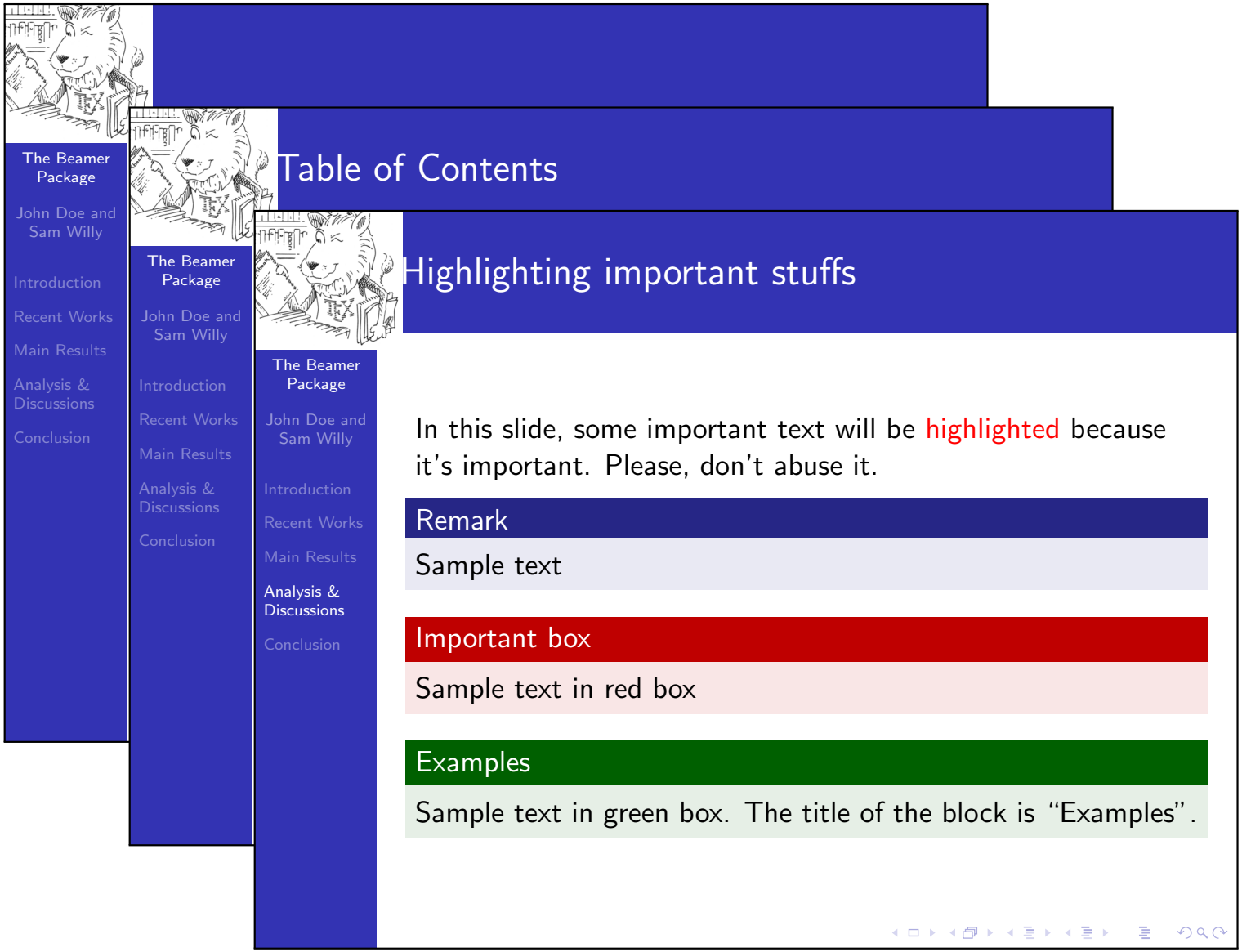
2 Recent Works

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Beamer Theme: Berkeley



Beamer Theme: Copenhagen

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Beamer Theme: Copenhagen

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Beamer Theme: Copenhagen

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Sample text in green box. The title of the block is "Examples".



Using a colortheme

A theme can be combined with a colortheme to change the color used for different elements:

```
\documentclass{beamer}
\usetheme{Madrid}
\usecolortheme{beaver}

\title{The Beamer Package}
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\author{John Doe\inst{1} and Sam Willy\inst{2}}
\institute{
\inst{1}%
Faculty of Mathematics\\A Famous University
\and
\inst{2}%
Faculty of Computer Science\\Another Famous University
}
\date{25 December 2023}
\logo{\includegraphics[height=2cm]{figures/ctan_lion.png}}
\begin{document}
\begin{frame}
\maketitle
```

Madrid THEME + beaver COLORTHEME

The Beamer Package

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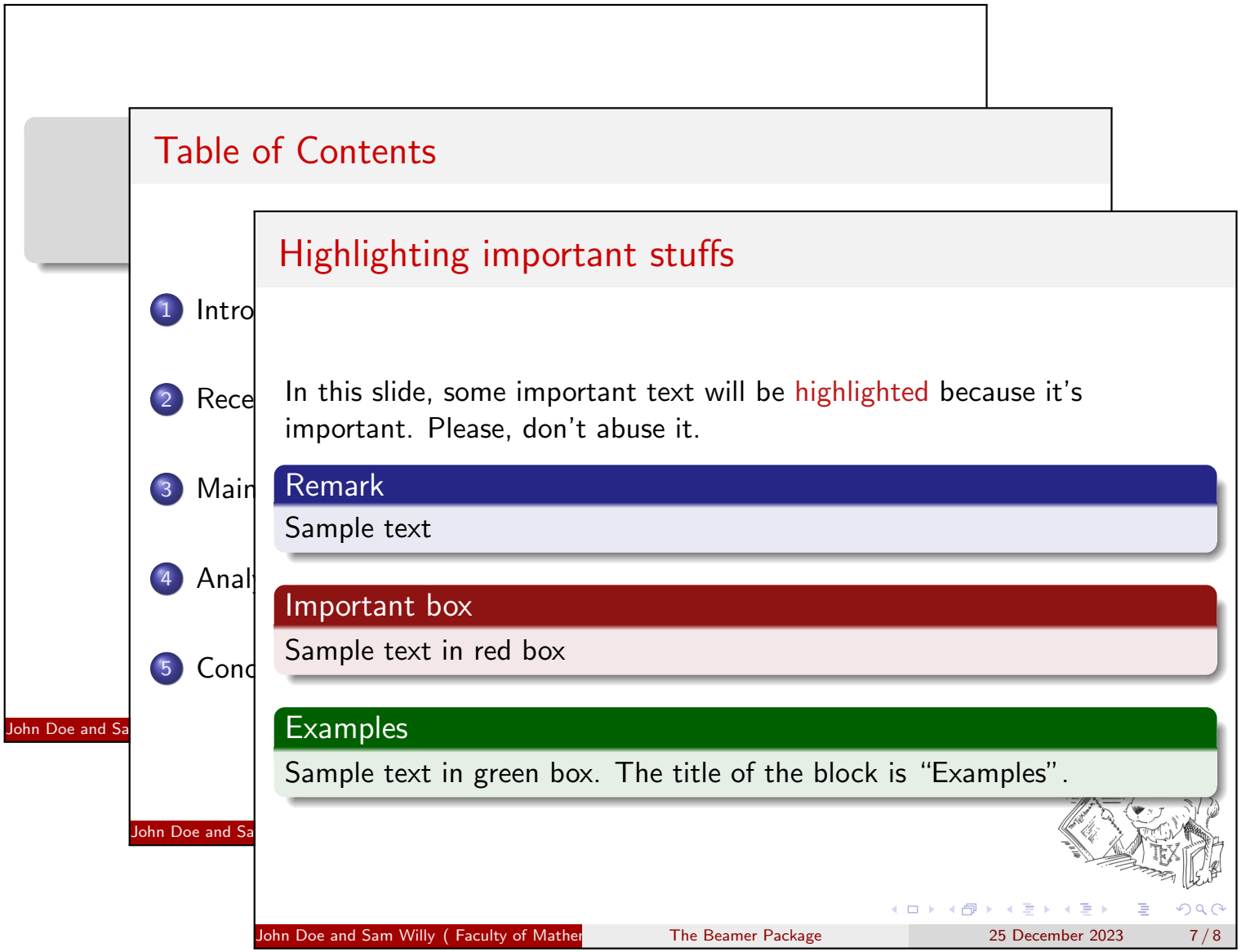
Madrid THEME + beaver COLORTHEME

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Madrid THEME + beaver COLORTHEME



Madrid + wolverine

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Madrid + wolverine

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Madrid + wolverine

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John Doe and Sa

John Doe and Sa

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Sample text in green box. The title of the block is "Examples".



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Berkeley + beaver



The Beamer Package

Creating good looking presentations in LaTeX

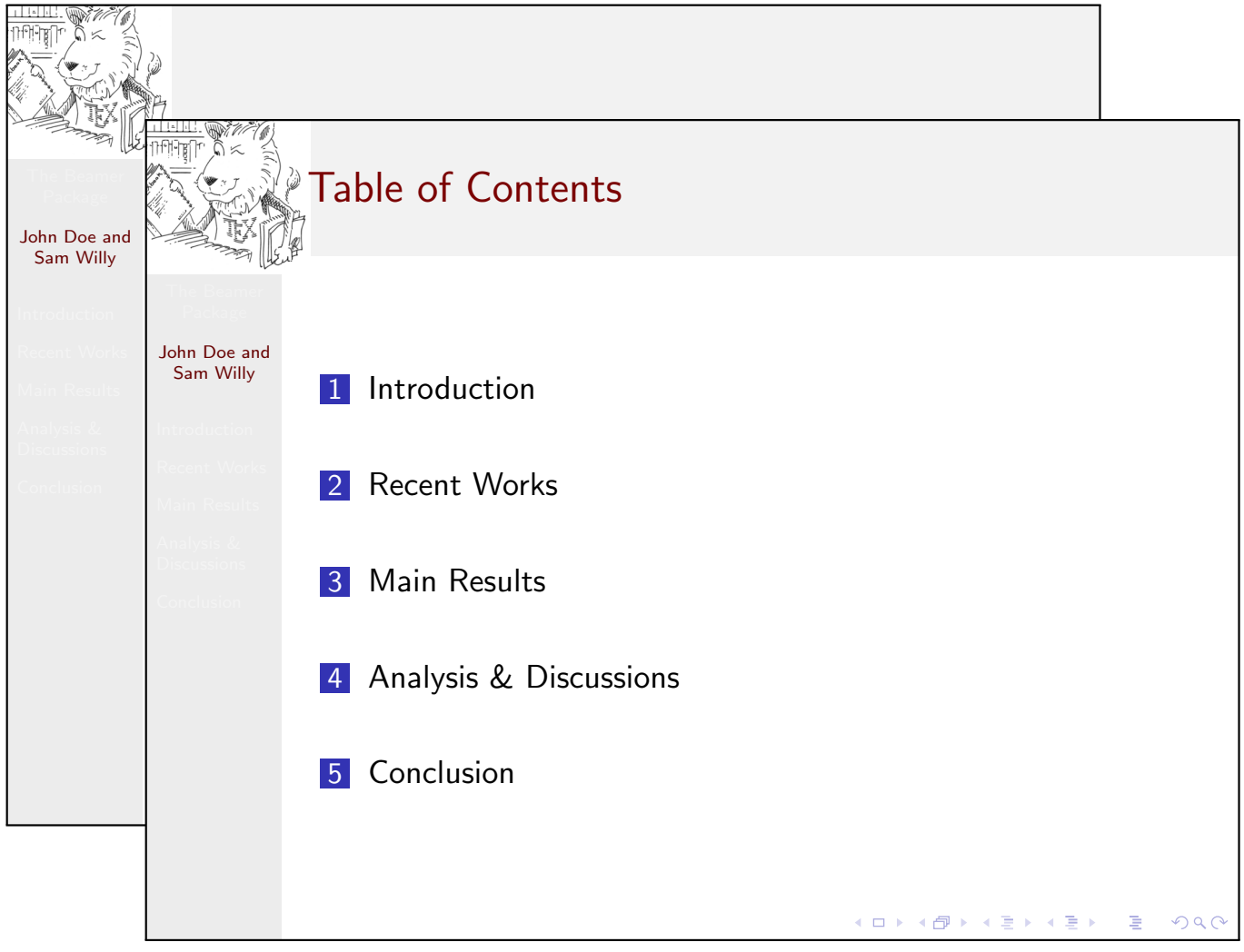
John Doe[1] and Sam Willy[2]

[1]Faculty of Mathematics
A Famous University

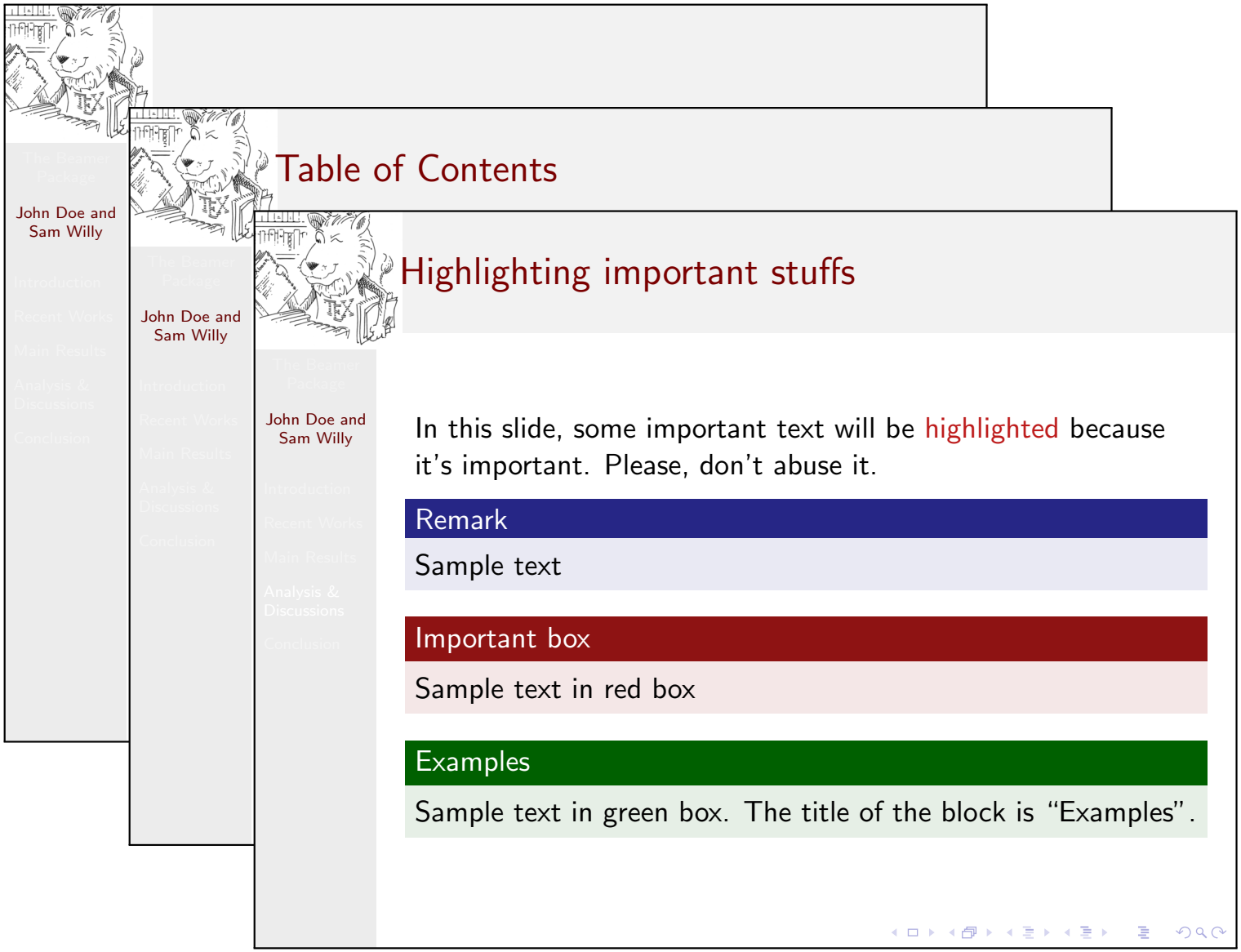
[2]Faculty of Computer Science
Another Famous University

25 December 2023

Berkeley + beaver



Berkeley + beaver



Themes and colorthemes

For a beamer presentation, we can use different combinations of themes and colorthemes. To have a complete list of them, see the following.

- The package documentation: `https://ctan.org/pkg/beamer?lang=en`.
- Piping's beamer theme matrix: `https://hartwork.org/beamer-theme-matrix/`.
- Petroff's beamer theme matrix: `https://mpetroff.net/files/beamer-theme-matrix/`

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RMUTT, November 2026

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